

Experiment 2: Monoalphabetic Substitution Cipher Monoalphabetic Substitution Cipher

Program:

```
#include <stdio.h> #include <ctype.h> #include <string.h>

void encrypt(char *message, char *key)

{ for (int i = 0; message[i] != '\0'; i++) { char ch = message[i];

    if (isalpha(ch)) {
        int index = toupper(ch) - 'A';
        char encrypted = key[index];

        if (islower(ch))
            encrypted = tolower(encrypted);

        message[i] = encrypted;
    }
}

}

void decrypt(char *message, char *key) { for (int i = 0; message[i] != '\0'; i++) { char ch =
message[i];

    if (isalpha(ch)) {
        char upper = toupper(ch);
        int index = -1;

        for (int j = 0; j < 26; j++) {
            if (key[j] == upper) {
                index = j;
                break;
            }
        }
    }

    if (index != -1) {
```

```

        char decrypted = 'A' + index;

        if (islower(ch))
            decrypted = tolower(decrypted);

        message[i] = decrypted;
    }
}

}

int main() { char message[1000]; char key[27];

printf("Enter a message: ");
fgets(message, sizeof(message), stdin);

printf("Enter 26-letter substitution key: ");
scanf("%26s", key);

for (int i = 0; i < 26; i++)
    key[i] = toupper(key[i]);

if (strlen(key) != 26) {
    printf("Invalid key! Key must contain 26 letters.\n");
    return 1;
}

for (int i = 0; i < 26; i++) {
    for (int j = i + 1; j < 26; j++) {
        if (key[i] == key[j]) {
            printf("Invalid key! Letters must be unique.\n");
            return 1;
        }
    }
}

encrypt(message, key);

```

```
printf("Encrypted message: %s", message);

decrypt(message, key);
printf("Decrypted message: %s", message);

return 0;

}
```

Input and Output :

```
Enter a message: HELLO
Enter 26-letter substitution key: QWERTYUIOPASDFGHJKLZXCVBNM
Encrypted message: ITSSG
Decrypted message: HELLO
```